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1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements a method of visualizing a pdf using kernel functions. This notebook draws random samples from a Student-t distribution and seeks to approximate a Gaussian distribution. The approximate pdf is visualized using a sum of weighted kernel "bump" functions. The distance between the reconstructed pdf approximation and the true pdf is computed via a Hellinger distance.

1 / 1 point

Run the code without any modifications. What is the Hellinger distance between the approximated pdf and the true pdf? Enter your answer with four digits to the right of the decimal point.

Note: Make sure that the code is configured with the default values of `width = 0.2`, `nSamples = 500`, and `sigma = 1` if you have changed them to experiment. Make sure that the Gaussian kernel function is uncommented and that the other two kernel functions are commented.

Also make sure that the code executes `randn("state", 0)` and `rand("state", 0)` and `randg("state", 0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

0.0129

✔ Correct
Yes. That is correct.

2. For the same Coursera Lab Jupyter notebook, change the value of `nSamples` to `nSamples = 5000` and run the code. What is the Hellinger distance between the approximated pdf and the true pdf? Enter your answer with four digits to the right of the decimal point.

1 / 1 point

Note: Make sure that the code is configured with the default values of `width = 0.2` and `sigma = 1` if you have changed them to experiment. Make sure that the Gaussian kernel function is uncommented and that the other two kernel functions are commented.

Also make sure that the code executes `randn("state", 0)` and `rand("state", 0)` and `randg("state", 0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

0.0004

✔ Correct
Yes, that is correct.

3. For the same Coursera Lab Jupyter notebook, reset the value of `nSamples` to `nSamples = 500`. Next, change the kernel function to implement the bi-weight kernel instead of the Gaussian kernel. (Uncomment the line that defines the bi-weight `KernelFunction` and comment the line that defines the Gaussian `KernelFunction`.)

1 / 1 point

Run the code. What is the Hellinger distance between the approximated pdf and the true pdf? Enter your answer with four digits to the right of the decimal point.

Note: Make sure that the code is configured with the default values of `width = 0.2` and `sigma = 1` if you have changed them to experiment. Also make sure that the code executes `randn("state", 0)` and `rand("state", 0)` and `randg("state", 0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

0.0157

✔ Correct
Yes, that is correct.